# 1. Clinical Question & Core Hypothesis

## Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

## Working Hypothesis

Primary hypothesis (ovarian serous carcinoma): MUC16 overexpression drives increased shedding of CA-125 into serum, and higher CA-125 levels reflect greater tumor burden and predict poorer OS and PFS as well as reduced chemotherapy responsiveness. This relationship is mediated by mechanistic edges such as MUC16–mesothelin–cell adhesion interactions and activation of pro-survival signaling pathways (e.g., PI3K/AKT, MAPK), which together create a causal path from MUC16/CA-125 expression in tumor tissue to adverse clinical outcomes. Secondary hypothesis (prostate adenocarcinoma): CA-125/MUC16 are present in a subset of prostate cancers and, when detectable, higher CA-125 correlates with increased tumor burden and worse prognosis, but the underlying edges are weaker and more context-dependent, likely reflecting shared inflammatory/immune-modulatory pathways rather than direct MUC16-driven adhesion or shedding. This framework will be tested by constructing constrained-path networks (3–4 hops) using edges labeled as expressed in/overexpressed in, biomarker for/serum marker, cleaved → soluble, associated with prognosis (OS/PFS), predicts therapy response, and binds/interacts with, ensuring provenance with PMID or database sources. The two parallel Tier-1 axes will enable a hierarchical triage to separate robust ovarian mechanistic signals from weaker prostate-context signals.

# 2. Why This Gene — Not Others

# 3. Clinically Testable Predictions

**Prediction 1:** retrospective analyses using OV and PRAD cohorts to correlate tissue MUC16 and serum CA-125 with OS/PFS and therapy response, adjusting for stage;

**Prediction 2:** in vitro manipulation of MUC16 in ovarian cancer cell lines to assess adhesion (mesothelial models), invasion, and chemosensitivity, with concomitant measurement of CA-125 release and signaling pathway activity;

**Prediction 3:** in vivo xenograft models to evaluate tumor burden, CA-125 shedding, and treatment response;

# 4. Minimum Viable Validation Plan

## A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

## B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

## C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

## Alternative Directions